

# Abstractions of Behavior

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## I. ABSTRACTIONS OF BEHAVIOR

In the given scenario, you are currently physically in China, and you are controlling a humanoid robot in Brazil to play soccer using your thoughts based on the advancements of brain computer interfaces. In this example, you can split the responsibilities of the robot's behavior into different levels to see who or what is technically responsible for each factor. As a whole, it can be determined that the responsibility of the robot would be split amongst the user and the robot itself [1]. Looking at the overarching goal or task level of what the action entails, the user would typically be in control and commanding the lead. For example, playing the game and making major decisions would ultimately be left to the user. Actions such as running, walking, dribbling, or shooting are all actions that are left to the discretion of the human and are completely detached from the robot. The next level that we will look at would be micro commands that occur within the robot during more high level goals. This would be actions such as dribbling, where the micro commands would be like beginning dribbling with your left foot or speed up your dribbling for the next 10 feet. The responsibilities for these actions are split between both the user and the robot. The user would be the one determining the actions while the robot would be loading predefined behaviors while modifying the mechanism based on human inputs. The lowest level would be completely controlled by the robot. These specified actions would be the equivalent to subconscious actions within the body, but now these actions would be within the robot instead. Things like balancing, temperature control, and joint movements would be completely independent from the human actions controlled by the user. These layers distribute the roles across different levels for the user and robot, leading to a shared responsibility in controlling the behavior of the robot.

## II. LEVELS OF INDIRECTION

With a change in context, the robot is now not being controlled using thought, but completely via mimesis, the responsibilities are unchanged. In this situation, if the robot shadows the user's own movements, the user's body becomes the control and source of the signals for the robot. Unlike the previous example, in which depending on the layer of command looked at, the responsibility of control is shared between the user and the robot. The user's bodily movements is what defines the intentions and tasks of the robot. Although the robot is still in control of more subconscious tasks and lower level functionalities, the user itself has responsibilities over

the robot with higher level tasks. The robot is what interprets and executes the signals given to the robot through the user's action. In this case it is simply the difference between thought communication and reading bodily movements in real time. Although there is a shift in modality of the control signal for the robot, the distribution of responsibilities do not change.

If in the case that instead of mimesis, the robot is now controlled using a game controller instead, the situation does not change much. Similarly to both previous situations, the responsibilities remain the same. The human component issues the commands through the controller to the robot. The robot would then interpret the signals and execute the commands. Each level of responsibility would remain identical to the previous situations, since the only change would be the modality of communication between the user and the robot [2]. The controller itself would not be an entity responsible for the behavior of the robot, since the controller would only be an instrument in which the human input commands are mapped onto the controller so that the robot can interpret the signals.

## III. BOUNDARIES OF INTELLIGENCE

To determine what parts of the casual chain of events in the behavior of the robot can be intelligent, I will be defining intelligence based on whether or not the entity is able to learn, adapt, and make intentions within decisions based on different situations [3]. In the given scenario, the user's interaction with the robot is through different mediums such as the mimesis, brain computer interface, or the game controller. The user's own brain decides the input commands and gives the actions mapped through the medium to the robot, where the robot interprets the signal and performs the action. The intelligence of this process comes completely through the user since the user is the one who sets the goals, makes the decisions, and adapts to the situation. No matter what the robot does, it all comes back to the user's goals and commands that is sent to the robot. This confirms that the user is responsible for the intelligence behavior behind the soccer game.

Although the most intelligent aspect of the behavior comes from the user, the robot itself also indicates some levels of intelligence from its multitude of automatic functionalities. When it comes to non user command actions that the robot must take to properly perform user inputted commands, the robot can demonstrate a high level of intelligence. For example, the robot knows how to balance, dribble, and perform micro actions all without human intervention. This confirms that the robot does have some smart elements within it. The

robot and the user both contribute to the intelligence of the behavior of the robot, just at different levels and in different ways.

#### REFERENCES

- [1] M. A. Arbib, *The Metaphorical Brain 2: Neural Networks and Beyond*, Wiley-Blackwell, 1989.
- [2] R. A. Brooks, "Intelligence without representation," *Artificial Intelligence*, vol. 47, no. 1–3, pp. 139–159, 1991.
- [3] S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson, 2021.